

SovereignShield

1. The Decision

An independent Azure-native reference study in synthetic-first statistical platform delivery.

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Consider a bounded pilot when an organization needs external specialists to implement a governed data workflow without giving them confidential production records. Provide an approved synthetic schema, sample lifecycle cases and an explicit access matrix. Retain production deployment, data-release and risk-acceptance decisions with the institution.

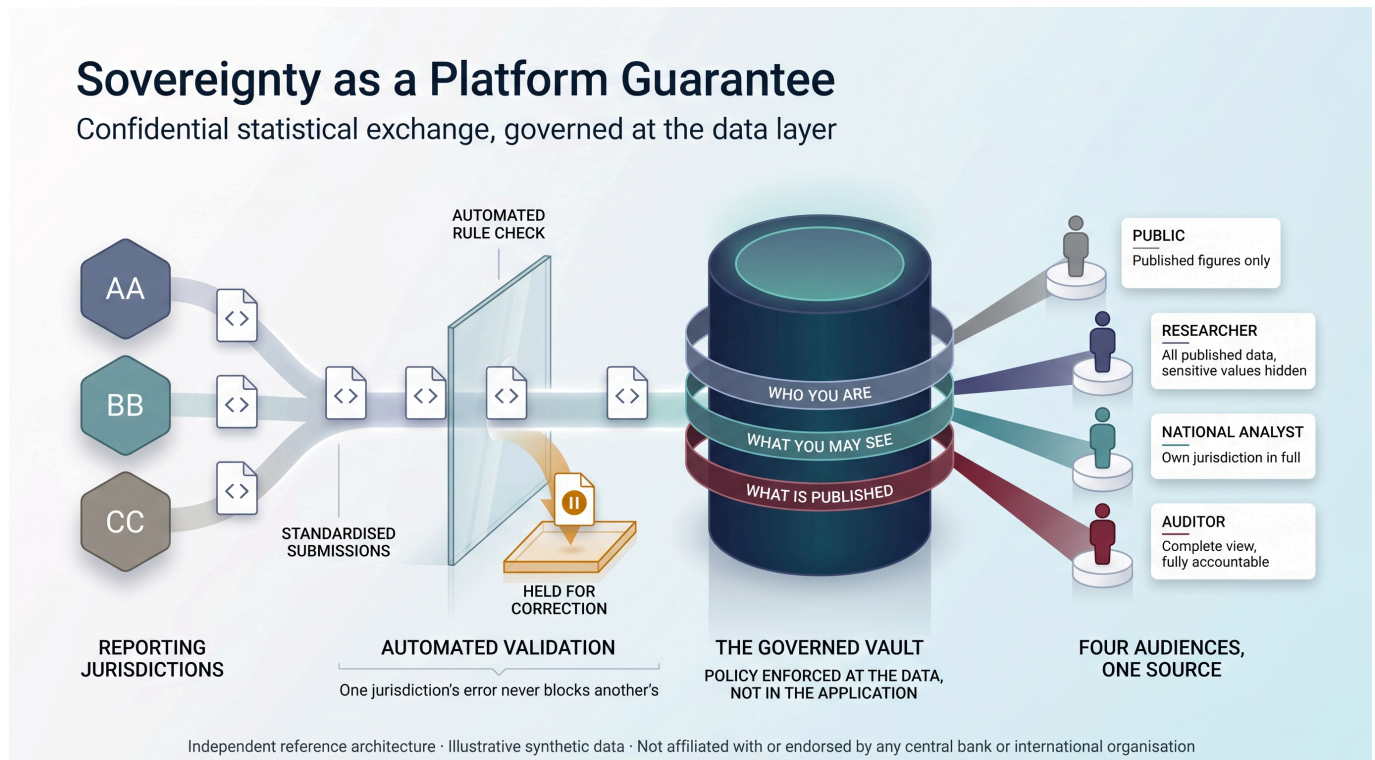
The decision is whether this delivery pattern reduces engagement and handover friction for a specific workload. It is not whether a synthetic demonstration proves production readiness.

2. Who Gets What

Persona	Visible Observations	Measure Access
Public	Current published, explicitly free	Free values only
Researcher	Current published structure across countries	Restricted values withheld unless separately entitled
Regional submitter	Own published/rejected filings plus public foreign observations	Own values in full; public foreign values
Administrator	All countries and lifecycle history	Full access under privileged governance

Researcher discovery creates a basis for requesting an agreement with the originating country. It does not confer permission to access a restricted measure. Multiple memberships combine explicitly; offboarding must remove every relevant entitlement and control-plane right.

3. One Architecture



Synthetic national submissions enter a protected archive. Validation makes a country/period verdict. Delta history separates accepted state from rejected arrivals. Unity Catalog evaluates query-time row and column policies. A gateway serves current SDMX feeds and distinct audit CSVs.

The image describes the architecture, not proof of physical country residency or complete confidentiality protection. The demonstrator uses a shared workspace with logical jurisdiction boundaries.

4. A Revision That Matters

An accepted CA filing contains two observations: **1.111** and **2.222**. A later accepted full replacement contains only the first, revised to **3.333**.

The replacement closes both previous current records and inserts the new snapshot in one Delta transaction. The omitted observation becomes non-current, while unrelated US observations and other periods remain unchanged.

A rejected replacement does not close accepted records. Its rows retain the submission ID, submitted/received timestamps, failed observation rule and batch rejection reason. Replaying that message adds no duplicate; a genuinely new identical message remains a distinct filing.

Three decimal places are this demonstration's explicit profile, not a universal SDMX rule. Original arrivals preserve the source evidence.

5. Evidence and Limits

Demonstrated Locally	Still Required
Failure-injection policy tests and no-detach deployment path	Actual UC migration and policy-binding acceptance
Decimal fidelity, code rejection, official JSON Schema checks	Full provisioning/content constraints and independent rule review
Real local Delta atomic commits, replay and smaller replacement cases	Distributed conflicts, recovery and production-volume measurements
Persona mirror, API workflows, desktop/mobile portal checks	Live signed-in persona matrix and two-host release verification

RLS and masking do not stop statistical inference from public totals. A production release needs an approved disclosure-control method, not simply stronger wording about Zero Trust.

Detailed methods and unresolved gates are in [Release Evidence](#). Historical screenshots are labelled historical, not fresh release results.

6. Operating Ownership

Data authority: Owns confidentiality classification, allowable use, secondary suppression, sender contracts and release approval.

Platform owner: Owns service principals, policies, deployment, backups, observability, access reviews and incident response. Resources must not depend on the contractor's personal identity.

External specialist: Builds against synthetic contracts, supplies executable tests and migration evidence, and hands over code and operational knowledge without requiring standing production access.

Independent reviewer: Approves promotion and residual risk. PR tests carry no cloud credentials; privileged plans and deployments use a protected manual gate.

Full offboarding reviews groups, sessions, tokens, Azure/vault/GitHub rights and delegated ownership. A group removal is one control, not the whole process.

7. Investment and Roadmap

Discovery: Agree the synthetic contract, access matrix, snapshot semantics, residency requirements and evaluation measures. Compare retaining the current system, extending established SDmx tooling, and an Azure-native pilot.

Bounded pilot: Fund implementation and independent control review within an agreed time and spending ceiling. Default to single-node evaluation; larger compute and Photon require a separate cost decision. Measure onboarding effort, time to accepted/rejected feedback, replay recovery, query latency, and cost per accepted submission.

Assurance: Rehearse protected migration, test the live persona matrix, recovery and offboarding, validate disclosure control, and obtain legal/provenance clearance.

Operational adoption: Proceed only when the institution accepts the evidence, operating model and recurring cost. Estimate Azure resources explicitly, then compare tagged actual costs; do not invent savings or extrapolate laptop benchmarks into a production business case.

8. What to Ask Next

Which access and revision cases would your architecture board require before permitting a pilot? Can an external contributor reproduce them without production data? Who accepts the statistical disclosure risk? Who owns the delivered runtime after the engagement ends?

The contribution is a transparent integration and delivery example, complementing SDMX registries, reference infrastructure and pysdmx. It is not a claim that other organizations lack these controls or that this architecture is globally novel.

[Repository](#) | [Full whitepaper](#) | [Evidence and migration](#)

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